

Given: Due:	Further Maths (Core Pure) HW 18	Name: MARK SCHEME
Chapter 3: Series		
Q1.		
(a) Use the standard results for $\sum_{r=1}^n r^3$ and $\sum_{r=1}^n r$ to show that		
$\sum_{r=1}^n (r^3 + 6r - 3) = \frac{1}{4}n^2(n^2 + 2n + 13)$		
for all positive integers n .		
(5)		
(b) Hence find the exact value of		
$\sum_{r=16}^{30} (r^3 + 6r - 3)$		
(2)		

In part (a) many candidates could start correctly by splitting the sum into three parts and substituted appropriate expressions for each sum. Those with any mistakes at this stage were unable to score any further marks. The subsequent algebra defeated some, and quite a few candidates could not establish the

printed result. There were very few candidates who thought $\sum_{r=1}^n 3 = 3$. Part (b) was usually answered correctly by the majority of candidates but a significant number calculated $S_{30} - S_{16}$. A small minority substituted into the original cubic to find the sum.

Most candidates gained the first three marks for use of the summation formula. The anticipated error of -1 rather than $-n$ for the final term was rarely seen. The best way to proceed was to cancel the numerical parts expand and factorise to obtain $n(n^2 + 5n + 2)$ and then factorise to the given answer. Some candidates made the question more difficult by keeping fractions, but most were successful in obtaining the required expression. Part (b) was accessible to all and was usually correct. Occasionally $S_{20} - S_{11}$ was seen and a few candidates substituted 20 and 10 into the $6r^2 + 4r - 1$.

Question Number	Scheme	Notes	Marks
(a)	$\sum_{r=1}^n (r^3 + 6r - 3)$		
	$= \frac{1}{4}n^2(n+1)^2 + 6 \cdot \frac{1}{2}n(n+1) - 3n$	M1: An attempt to use at least one of the standard formulae correctly in summing at least 2 terms of $r^3 + 6r - 3$	M1A1B1
		A1: Correct underlined expression.	
		B1: $-3 \rightarrow -3n$	
	$= \frac{1}{4}n^2(n+1)^2 + 3n^2 + 3n - 3n$		
	If any marks have been lost, no further marks are available in part (a)		
	$= \frac{1}{4}n^2(n+1)^2 + 3n^2$ $= \frac{1}{4}n^2((n+1)^2 + 12)$	Cancels out the $3n$ and attempts to factorise out at least $\frac{1}{4}n$.	dM1
	$= \frac{1}{4}n^2(n^2 + 2n + 13)$ (AG)	Correct answer with no errors seen.	A1 *

Provided the first 3 marks are scored, allow the next two marks for correctly showing the algebraic equivalence. E.g. showing that both $\frac{1}{4}n^2(n+1)^2 + 6\frac{1}{2}n(n+1) - 3n$ and $\frac{1}{4}n^2(n^2 + 2n + 13) = \frac{1}{4}n^4 + \frac{1}{2}n^3 + \frac{13}{4}n^2$ There are no marks for proof by induction but apply the scheme if necessary.		
		[5]
(b)	$S_n = \sum_{r=16}^{30} (r^3 + 6r - 3) = S_{30} - S_{15}$ $= \frac{1}{4}(30)^2(30^2 + 2(30) + 13) - \frac{1}{4}(15)^2(15^2 + 2(15) + 13)$ Use of $S_{30} - S_{15}$ or $S_{30} - S_{16}$ M1 NB They must be using $S_n = \frac{1}{4}n^2(n^2 + 2n + 13)$ not $S_n = n^3 + 6n - 3$ $= 218925 - 15075$ $= 203850$ 203850 A1 cao NB $S_{30} - S_{16} = 218925 - 19264 = 199661$ (Scores M1 A0)	
		[2]
		7 marks

Q2.

(a) Show, using the formulae for $\sum r$ and $\sum r^2$, that

$$\sum_{r=1}^n (6r^2 + 4r - 1) = n(n+2)(2n+1)$$

(5)

(b) Hence, or otherwise, find the value of $\sum_{r=1}^{20} (6r^2 + 4r - 1)$.

(2)

The majority of students answered part (a) correctly. Errors were mainly due to not replacing all the n 's with $3n$. The first was usually correct so the most common error seen was $\frac{1}{6}(3n)(n+1)(2n+1)$.

Very many fully correct solutions were seen from students in part (b) who were successful in part (a). Some students made life difficult by failing to take a factor of $\frac{n}{6}$ out at the start of their work; they should be encouraged to use the given solution as a guide to finding common factors and deal with these straight away. Students were usually able to expand each term to obtain quadratic expressions, this being done to a high degree of accuracy. A few errors were made with coefficients, but the main cause of any errors was that of making a sign error when subtracting the second term.

Question Number	Scheme	Marks
(a)	$6\sum r^2 + 4\sum r - \sum 1 = 6\frac{n}{6}(n+1)(2n+1) + 4\frac{n}{2}(n+1) - n$ $= \frac{n}{6}(12n^2 + 18n + 6 + 12n + 12 - 6)$ or $n(n+1)(2n+1) + (2n+1)n$ $= \frac{n}{6}(12n^2 + 30n + 12) = n(2n^2 + 5n + 2) = n(n+2)(2n+1)$ *	M1 A1, B1 M1 A1 (5)
(b)	$\sum_{r=1}^{20} (6r^2 + 4r - 1) - \sum_{r=1}^{10} (6r^2 + 4r - 1) = 20 \times 22 \times 41 - 10 \times 12 \times 21$ $= 15520$	M1 A1 (2) [7]

Q3.

(a) Using the formula for $\sum_{r=1}^n r^2$ write down, in terms of n only, an expression for

$$\sum_{r=1}^{3n} r^2$$

(b) Show that, for all integers n , where $n > 0$

$$\sum_{r=2n+1}^{3n} r^2 = \frac{n}{6}(an^2 + bn + c)$$

where the values of the constants a , b and c are to be found.

(4)

Most students used the general formulae successfully in part (a). Many students multiplied out all the brackets, only removing a factor at the end of their working and this was usually a successful approach. Of those removing a factor of $n/6$ or $n/3$ at the start, the biggest problem was carelessness in their working, leading to unnecessary slips.

Most students were fully successful in part (b), although those who had been unable to complete part (a) did not often attempt it. Common errors included inaccurate multiplying out of brackets, not using the formula shown in part (a), or calculating $f(n+1) - f(n)$ or other incorrect combinations.

Question Number	Scheme	Marks
(a)	$\sum_{r=1}^{3n} r^2 = \frac{1}{6} 3n(3n+1)(6n+1)$ or $\sum_{r=1}^{3n} r^2 = \frac{1}{2} n(3n+1)(6n+1)$ or equivalent	B1
(b)	See $\sum_{r=1}^{2n} r^2 = \frac{1}{3} n(2n+1)(4n+1)$ or equivalent	B1
	Attempt to use $\sum_{r=1}^{3n} r^2 - \sum_{r=1}^{2n} r^2 = \frac{n}{6} \{3(3n+1)(6n+1) - 2(2n+1)(4n+1)\}$	M1
	$= \frac{n}{6} \{(54n^2 + 27n + 3) - (16n^2 + 12n + 2)\}$	dM1
	$= \frac{n}{6} \{(38n^2 + 15n + 1)\}$	A1
	$(a = 38, b = 15, c = 1)$	
		(4)
		5 marks
Notes		
(a) B1: Either right hand side or exact equivalent - isw if expanded		
(b) B1: States or uses $\sum_{r=1}^{2n} r^2 = \frac{1}{3} n(2n+1)(4n+1)$		
M1: Subtracts their sum to $2n$ or $2n-1$ and attempts to factorise by $\frac{n}{6}$ seen anywhere.		
dM1: Expands two quadratics dependent on first M1		
A1: cao		

Q4.

(a) Using the formulae for $\sum_{r=1}^n r$ and $\sum_{r=1}^n r^2$, show that

$$\sum_{r=1}^n (r+1)(r+4) = \frac{n}{3}(n+4)(n+5)$$

for all positive integers n .

(5)

(b) Hence show that

$$\sum_{r=n+1}^{2n} (r+1)(r+4) = \frac{n}{3}(n+1)(an+b)$$

where a and b are integers to be found.

(3)

In part (a) virtually all candidates expanded correctly and substituted the standard formulae as well as identifying the "+ n". Weaker candidates then often struggled with the resulting algebra. Candidates should be encouraged to use the printed result to identify potential factors if possible, rather than multiply out completely and then start working towards the result.

Part (b) was met with less success with a large number of candidates starting again from scratch by expanding the brackets rather than using the hence. Those who did use the result in (a) often misinterpreted what was meant by S_{3n} and this sum often ended up as $3S_n$.

Question Number	Scheme	Marks
(a)	$\sum_{r=1}^n (r+1)(r+4)$ $= \sum_{r=1}^n r^2 + 5r + 4$ $= \frac{n}{6}(n+1)(2n+1) + 5\frac{n}{2}(n+1) + 4n$ $= \frac{n}{6}\{(n+1)(2n+1) + 15(n+1) + 24\}$ $= \frac{n}{6}\{2n^2 + 3n + 1 + 15n + 15 + 24\}$ $= \frac{n}{6}(2n^2 + 18n + 40) \text{ or } = \frac{n}{3}(n^2 + 9n + 20)$ $= \frac{n}{3}(n+4)(n+5) \text{ ** given answer**}$	B1 M1 A1 dM1 A1* (5)
(b)	$\sum_{r=n+1}^{2n} (r+1)(r+4) = \frac{2n}{3}(2n+4)(2n+5) - \frac{n}{3}(n+4)(n+5)$ $= \frac{n}{3}\{8n^2 + 36n + 40 - n^2 - 9n - 20\}$ $= \frac{n}{3}\{7n^2 + 27n + 20\} = \frac{n}{3}(n+1)(7n+20) \text{ or } a=7, b=20$	M1 dM1 A1 (3)
Notes (a) B1: Expands bracket correctly to $r^2 + 5r + 4$ M1: Uses $\frac{n}{6}(n+1)(2n+1)$ or $\frac{n}{2}(n+1)$ correctly. A1: Completely correct expression. dM1: Attempts to remove factor $\frac{n}{6}$ or $\frac{n}{3}$ to obtain a quadratic factor. Need not be 3 term. A1: Completely correct work including a step with a collected 3 term quadratic prior in the bracket with correct printed answer. Accept approach which starts with LHS and then RHS which meet at $\frac{n^3}{3} + 3n^2 + \frac{20n}{3}$. Award marks as above. NB If induction attempted then typically this may only score the first B1. However, consider the solution carefully and award as above if seen in the body of the induction attempt. (b) M1: Uses $f(2n) - f(n)$ or $f(2n) - f(n+1)$ correctly. Require all 3 terms in $2n$ (and $n+1$ if used). dM1: Attempts to remove factor $\frac{n}{6}$ or $\frac{n}{3}$ to obtain a quadratic factor. Need not be 3 term. A1: Either in expression or as above.		

Q5.

(a) Use the results for $\sum_{r=1}^n r$ and $\sum_{r=1}^n r^2$ to show that

$$\sum_{r=1}^n (2r-1)^2 = \frac{1}{3}n(2n+1)(2n-1)$$

for all positive integers n .

(6)

(b) Hence show that

$$\sum_{r=n+1}^{3n} (2r-1)^2 = \frac{2}{3}n(an^2 + b)$$

where a and b are integers to be found.

Question Number	Scheme	Notes	Marks
(a)	$\{S_n\} = \sum_{r=1}^n (2r-1)^2$		
	$= \sum_{r=1}^n 4r^2 - 4r + 1$	Multiplying out brackets and an attempt to use at least one of the two standard formulae correctly.	M1
	$= 4 \cdot \frac{1}{6} n(n+1)(2n+1) - 4 \cdot \frac{1}{2} n(n+1) + n$	First two terms correct. + n	A1 B1
	$= \frac{2}{3} n(n+1)(2n+1) - 2n(n+1) + n$		
	$= \frac{1}{3} n \{ 2(n+1)(2n+1) - 6(n+1) + 3 \}$	Attempt to factorise out $\frac{1}{3}n$ Correct expression with $\frac{1}{3}n$ factorised out with no errors seen.	M1 A1
	$= \frac{1}{3} n \{ 2(2n^2 + 3n + 1) - 6(n+1) + 3 \}$		
	$= \frac{1}{3} n \{ 4n^2 + 6n + 2 - 6n - 6 + 3 \}$		
	$= \frac{1}{3} n (4n^2 - 1)$		
	$= \frac{1}{3} n (2n+1)(2n-1)$	Correct proof. No errors seen.	A1 *
	Note that there are no marks for proof by induction.		(6)
(b)	$\sum_{r=n+1}^{3n} (2r-1)^2 = S_{3n} - S_n$		
	$= \frac{1}{3} \cdot 3n(6n+1)(6n-1) - \frac{1}{3} n(2n+1)(2n-1)$	Use of $S_{2n} - S_n$ or $S_{2n} - S_{n+1}$ with the result from (a) used at least once. Correct unsimplified expression. E.g. Allow $2(3n)$ for $6n$.	M1 A1
	Note that (b) says hence so they have to be using the result from (a)		
	$= n(36n^2 - 1) - \frac{1}{3} n(4n^2 - 1)$		
	$= \frac{1}{3} n(108n^2 - 3 - 4n^2 + 1)$	Factorising out $\frac{1}{3}n$ (or $\frac{2}{3}n$)	dM1
	$= \frac{1}{3} n(104n^2 - 2)$		
	$= \frac{2}{3} n(52n^2 - 1)$	$\frac{2}{3} n(52n^2 - 1)$	A1
	$\{a = 52, b = -1\}$		(4)
			10